

INDIAN LANGUAGES AND IT

As Hindi and regional languages become mainstream on computers, English literacy is no longer a requirement to be computer conversant

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Personal computers (PCs) made their appearance in India in 1985. At that time, English was the only language understood by computers. Nobody thought that computers could ever deal with Indian languages. Due to this, only people who understood English could use computers, which comprised mostly the urban population. In the course of time, there have been vast changes in the computer field and, gradually, Indian languages have made their presence felt on computers and on the Internet.

The first move

The first major effort in this direction was made by the Centre for Development of Advanced Computing (C-DAC) when it developed a graphic and intelligence-based script technology (GIST) card—a PC add-on card which allowed users to work in Devanagari and many other Indian languages in MS-DOS-based applications like dBase, FoxPro, Lotus 1-2-3 and WordStar.

In 1995, Microsoft released the Windows 95 operating system bundled with Internet Explorer 1.0, which had built-in support for a dial-up connection and TCP/IP. With this, the Internet became accessible to many more people in India. As the Internet started gaining popularity, people thought of being able to use Indian languages on the Internet, and this led to the development of fonts for Indian languages.

In 1997, some true type fonts were released, which could be used to type



the Devanagari script on Windows-based computers and on webpages. These fonts had some compatibility issues, and the manual downloading of fonts and their subsequent installation was necessary. To overcome this problem, dynamic fonts were designed that didn't require the physical download, and fonts were available when the webpage was viewed. But these dynamic fonts worked on some browsers only. To overcome this, eventually the Unicode system of fonts was invented. This has helped in the development of many Indian language fonts.

Unicode font support

Now, almost all products are available with Unicode font support. In the Unicode system, each character is given a unique number, and this number is independent of the operating sys-

tem and programming language. So the problem of font compatibility ceases to exist, and there is no need to download fonts and install them on your computer every time you visit a new website. The Unicode Consortium oversees a Common Locale Data Repository (CLDR) project, which provides key building blocks for software to support the world's languages. Data from this project is used by companies across the globe for both the internationalisation and localisation of their products.

The latest version of Unicode is 5.1 and has almost 100,000 characters, many of which have been added to the Indic script. The Unicode text segmentation algorithm has also improved a lot, which helps in the processing of Indic languages.

Unicode 5.1 has added around 1624 newly encoded characters that